Sports Medicine
https://doi.org/10.1007/s40279-021-01585-9

REVIEW ARTICLE

Evaluating the Effects of Increased Protein Intake on Muscle Strength, Hypertrophy and Power Adaptations with Concurrent Training: A Narrative Review

Donny M. Camera[1,2]

Accepted: 16 October 2021
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2021

Abstract

Concurrent training incorporates dual exercise modalities, typically resistance and aerobic-based exercise, either in a single session or as part of a periodized training program, that can promote muscle strength, mass, power/force and aerobic capacity adaptations for the purposes of sports performance or general health/wellbeing. Despite multiple health and exercise performance-related benefits, diminished muscle hypertrophy, strength and power have been reported with concurrent training compared to resistance training in isolation. Dietary protein is well-established to facilitate skeletal muscle growth, repair and regeneration during recovery from exercise. The degree to which increased protein intake can amplify adaptation responses with resistance exercise, and to a lesser extent aerobic exercise, has been highly studied. In contrast, much less focus has been directed toward the capacity for protein to enhance anabolic and metabolic responses with divergent contractile stimuli inherent to concurrent training and potentially negate interference in muscle strength, power and hypertrophy. This review consolidates available literature investigating increased protein intake on rates of muscle protein synthesis, hypertrophy, strength and force/power adaptations following acute and chronic concurrent training. Acute concurrent exercise studies provide evidence for the significant stimulation of myofibrillar protein synthesis with protein compared to placebo ingestion. High protein intake can also augment increases in lean mass with chronic concurrent training, although these increases do not appear to translate into further improvements in strength adaptations. Similarly, the available evidence indicates protein intake twice the recommended intake and beyond does not rescue decrements in selective aspects of muscle force and power production with concurrent training.

✉ Donny M. Camera
dcamera@swin.edu.au

[1] Department of Health and Biostatistics, Swinburne University, Melbourne, Australia

[2] Department of Health Science and Biostatistics, Swinburne University, Room SPW224; Mail H21, PO Box 218, Hawthorn M., Victoria 3122, Australia

Published online: 25 November 2021

Key Points

Dietary protein is a critical substrate for providing amino acids to facilitate skeletal muscle repair and regeneration during recovery from exercise. However, it is not clear whether increased protein intake can enhance exercise-induced adaptation responses and subsequently mitigate potential interference in muscle strength, hypertrophy and force/power output observed with concurrent training.

The available literature indicates high protein intake stimulates acute rates of myofibrillar protein synthesis as well as augments significant increases in lean mass with concurrent training that may be beneficial for activities requiring increased muscle mass for performance.

Current evidence from longer duration studies indicates increased protein intake has limited capacity to attenuate interference in muscle strength and power adaptations with concurrent training compared to resistance training performed separately.

1 Introduction

Undertaking regular exercise is an established mediator of numerous physiological and psychological health benefits [1]. The specificity of training dictates that physiological and metabolic bodily adaptations after exercise training are dependent on the specific mode, volume and intensity of exercise training practiced [2, 3]. Exercise has been traditionally categorized into resistance or endurance/aerobic modalities that typically produce different, and at times, contrasting phenotypic responses from each other [4]. Endurance training (i.e., cycling, running) leads to increases in oxidative enzyme activity, intramuscular myoglobin, mitochondrial content via increased rates of mitochondrial protein synthesis for improved substrate metabolism [5], size and number of mitochondria [6, 7] and capacity to oxidize lipids [8]. Resistance training (i.e., machine based or free weights) stimulates increases in myofibrillar protein accretion responsible for muscle contraction and remodeling [9, 10], muscle force production and increased cross-sectional area of type II fibers [2, 11]. It must be noted that this exercise mode specificity in training adaptation exists on a continuum, with evidence of, for example, increases in skeletal muscle fiber size with aerobic-based exercise [12, 13] and increased skeletal muscle lipid oxidation capacity [14] and mitochondrial DNA copy number [15] following resistance training. More recently, high-intensity interval training (HIIT) has become a popular type of exercise strategy to increase overall health and well-being [16]. HIIT involves brief periods of high-intensity exercise (e.g., cycling) interspersed by low-intensity exercise/passive rest. HIIT has extensively been shown to improve skeletal muscle oxidative capacity, whole-body and skeletal muscle fatty acid oxidation, and maximal oxygen uptake (VO_{2max}) [17]. From an athletic performance perspective, HIIT is often implemented into training programs to emulate 'match situations' characteristic of high-intensity exertions in given activities [18, 19]. Interestingly, a single bout of HIIT can stimulate rates of myofibrillar protein synthesis [20] and promote increases in lean mass and strength [21], suggesting HIIT can also potentiate, albeit to a lesser degree than resistance training, anabolic adaptation responses in skeletal muscle.

Adult World Health Organization global recommendations for physical activity promoting overall health and well-being stipulate a combination of both resistance and aerobic-type exercises to improve cardiovascular and muscular fitness [22, 23]. The combined performance of both resistance and endurance exercise in a training program can be termed concurrent training and has been shown to simultaneously improve multiple indices of skeletal muscle and cardiovascular health including muscle strength, peak oxygen uptake (VO_{2peak}) and oxygen consumption [24, 25]. For the purposes of this review, concurrent exercise will be generically defined as the performance of resistance and aerobic-based exercise modalities interchangeably (i.e., in either sequence) in a single training session and/or when both types of exercise modalities are performed on different occasions that form part of a periodized training program [26, 27]. Many individual and team-based sports such as basketball, rugby, rowing and Australian Rules Football require a diverse combination of muscle strength, explosive power and aerobic capacity in addition to match-specific movements (e.g., jumping, tackling, changing directions and rapid acceleration) for successful performance [28, 29]. Moreover, specific field positions within a sport and level of competition can impose differing and unique physiological demands [30].

While concurrent training is a popular and required strategy among athletes and non-athletes alike for enhanced performance and health and also employed by rehabilitation clinicians following injury, surgery, etc., evidence from multiple studies demonstrates that the magnitude of increase with concurrent training in muscle strength, power and, to a lesser extent, hypertrophy, can be attenuated when compared to resistance training undertaken in isolation [31–35]. Coined as the 'interference effect' from early work in the 1980s [34] and confirmed by subsequent studies [35], it has been shown that when high volumes (> 4 sessions per week) and intensity (> 70% VO_{2max}) of endurance exercise

are performed in a concurrent resistance and endurance training program that gains in muscle strength, hypertrophy and power are significantly lower compared to when the same resistance training program is performed in isolation. It must be noted that this attenuated 'anabolic' response is not always observed [23, 36, 37], and possible factors for this disparity, including the sequence, timing, volume and mode of concurrent training, as well as individual training status, have been extensively discussed elsewhere [26, 27, 38]. Moreover, some studies have reported greater increases in muscle cross-sectional area and mean fiber area with concurrent training compared to single-mode resistance training [39, 40]. Regardless, the capacity for concurrent training to reduce muscle strength, hypertrophy and power adaptations is perhaps not that surprising given the divergent nature of both contractile stimuli and distinct underlying molecular responses governing adaptation responses with endurance and resistance exercise. In this regard, it is also worth noting that concurrent training can also induce profound detrimental effects on the quality and performance of endurance training sessions/performance [41], with attenuated cycling time-trial performance [9] and running time to exhaustion [7] observed several days following a single bout of resistance training. While possible interference in muscle strength, hypertrophy and power adaptations with concurrent training may be statistically small among recreationally active populations, such effects can still impact performance outcomes in elite and sub-elite individuals.

A crucial caveat to the multitude of study designs that have previously investigated cellular and physiological responses to concurrent training is a lack of nutritional intervention/control (i.e., studies performed in the fasted state or without post-exercise nutrition), which is a critical factor to enhancing exercise adaptation responses [42], particularly considering most individuals spend the majority of the day in a postprandial state [43]. Nutrient ingestion during post-exercise recovery is important to replenish endogenous substrate stores and to facilitate repair and remodeling of skeletal muscle. Protein and carbohydrate are widely studied macronutrients regarding their capacity to generate changes in exercise-induced adaptation and athletic performance. When considering the high energy demands potentially inherent with concurrent training due to the large training volumes (generally compared to single-mode training), it has widely been established that replenishing muscle energy stores and post-exercise rehydration is crucial for exercise performance. As a result, much emphasis has been directed toward suitable post-exercise carbohydrate ingestion protocols for promoting muscle glycogen resynthesis [44] and helping avoid a net negative energy balance, which can be detrimental for lean mass accrual [45, 46]. While carbohydrate is therefore a fuel source of great importance within a concurrent training schedule, intricate discussion regarding the types, amount and timing of carbohydrate ingestion is beyond the scope of this review and the reader is directed toward previously published work on this area [47, 48]. Dietary protein is a critical substrate for providing amino acids to facilitate skeletal muscle repair and regeneration during recovery from exercise. Protein intake in the form of supplements is also a popular method for increasing protein intake, as evidenced by protein powder sales of almost US$ 5 billion in 2015 alone [49]. While the degree to which protein ingestion can amplify adaptation responses with resistance exercise, and to a lesser extent endurance exercise, has been highly studied, much less is known about the capacity for protein ingestion to enhance anabolic and metabolic responses with concurrent training. Indeed, the most recent International Society of Sports Nutrition Position Stand for protein and exercise [50] only provides protein intake recommendations for resistance or endurance exercise performed in isolation and does not consider concurrent training. Little is also known about whether protein ingestion can enhance adaptation responses and potentially mitigate potential interference in muscle strength, hypertrophy and force/power output observed with concurrent training.

Individuals who routinely undertake resistance/strength or aerobic-based training in isolation have higher protein requirements compared to non-exercisers [51–53]. However, whether individuals who perform both modalities concurrently (particularly in the same day) require similar or even greater protein intake has not been thoroughly investigated. Such information would be of practical relevance for those that undertake high-volume concurrent training for performance or general wellbeing. It has also been proposed that protein supplementation/high protein intake may provide the greatest potential for ergogenic effects with exercise when participants are in negative nitrogen and/or energy balance because of decreased resting rates of muscle protein synthesis (MPS) and a heightened sensitivity of MPS to protein feeding [54, 55]. Such a situation is highly probable in a concurrent training program with potentially greater caloric demand when performing high volumes of divergent exercise modalities as part of a regular training schedule. The aim of this review is to consolidate the available evidence for protein ingestion to facilitate anabolic and muscle force/power adaptation responses with concurrent resistance and aerobic-based training. The capacity for protein to promote muscle remodeling processes and augment recovery from high training volumes/muscle damage will also be discussed. A greater understanding of how increased protein intake (i.e., above recommended daily allowances) can ameliorate key physiological responses with concurrent training may provide important information in applied environments to ultimately prescribe more effective nutritional recommendations within this training environment that may negate interference with key adaptation responses.

2 Effects of Protein Ingestion on Rates of Muscle Protein Synthesis with Acute Concurrent Exercise

Protein ingestion enhances acute rates of MPS that, over time, promotes the accrual of muscle proteins forming the basis for muscle hypertrophy [56, 57]. Dietary protein stimulates MPS by initiating the transfer and incorporation of amino acids from these dietary protein sources into skeletal muscle proteins. Muscle anabolism stimulated by protein feeding balances against the periods of net catabolism that occur between feedings to generally ensure muscle mass remains constant in young healthy adults. Thus, protein is a critical substrate for providing amino acids to aid in skeletal muscle repair and regeneration during recovery from exercise. The capacity for protein ingestion to stimulate rates of MPS with resistance training is well established [58, 59]. Evidence also exists for protein to exert positive effects on MPS following endurance-based exercise as well as HIIT [60–63]. Comparatively, fewer studies have investigated the effects of protein ingestion on acute rates of MPS responses following a single bout of concurrent exercise.

Initial work to investigate the capacity for protein ingestion to augment rates of MPS with concurrent exercise was undertaken in a study aimed primarily at investigating the influence of muscle glycogen availability to regulate MPS responses [64]. Such a question can bear practical relevance to individuals undertaking multiple high-intensity exercise bouts within the same day or with limited recovery periods where there is a high likelihood of commencing a training session with reduced/sub-optimal muscle glycogen availability. This study incorporated an afternoon one-legged cycling protocol to purposely deplete muscle glycogen from one leg so that comparisons could be made between 'low' and 'normal' glycogen legs (Table 1). Following this session, participants were provided a low carbohydrate evening meal (~ 1 g/kg body mass) to limit glycogen re-synthesis, and the next morning, following ~ 10 h recovery, performed a one-legged session of leg press before receiving either a protein-carbohydrate or placebo beverage. Rates of myofibrillar protein synthesis were higher in the protein compared to placebo condition, thus supporting the well-established capacity for whey protein to augment post-exercise MPS responses compared to a non-protein control. However, contrary to the authors' hypothesis, there were no differences in MPS responses post-resistance training between 'low' and 'normal' glycogen legs for the protein-carbohydrate or placebo groups [64]. While this may indicate a negligible role of muscle glycogen status to alter acute rates of MPS, it is possible that repeated bouts of resistance training performed with low/sub-optimal glycogen availability, akin to a weekly/fortnightly training program where multiple aerobic-based and resistance training sessions are undertaken, may be required to induce a level of metabolic perturbation capable of negatively modulating MPS. Intuitively, in such instances, increased protein intake throughout recovery would be paramount to rescuing the potential for attenuated rates of MPS. Furthermore, co-ingestion of protein with carbohydrate may also be crucial in these circumstances to prevent caloric deficit, which can attenuate rates of MPS and may thus be implicated in longer term interference in muscle hypertrophy adaptations with concurrent training. These potential effects on MPS responses with increased protein (and carbohydrate) intake following repeated concurrent exercise bouts that induce significant muscle glycogen depletion remain an area of future investigation.

Donges and colleagues implemented a cross-over design study to determine acute MPS responses following a single bout of either resistance or endurance exercise in isolation, or concurrent resistance and endurance exercise, in sedentary, middle-aged men (Table 1), with a 20-g whey protein isolate beverage consumed immediately after the cessation of exercise [65]. The first major finding was that concurrent exercise was able to simultaneously increase acute post-exercise (1–4 h) rates of myofibrillar and mitochondrial protein synthesis. Notably, these increases in myofibrillar protein synthesis rates were equivalent to increases observed with the resistance exercise-only condition despite the participants only performing half the actual resistance exercise volume (4 sets compared to 8) during the concurrent exercise bout. Similarly, the increases in rates of mitochondrial protein synthesis with the concurrent exercise session were comparable to the magnitude of increase when endurance exercise was performed in isolation even though the participants only performed 20 min (as opposed to 40 min) during the concurrent exercise condition. While these results advocate protein to support acute muscle growth and metabolic processes following concurrent training, it must be noted that the participant cohort were untrained, sedentary and unaccustomed to the imposed exercise contractile stimulus. Such factors are known to significantly influence the molecular signals responsible for generating proteins that are synthesized after an acute exercise session [5]. Moreover, two limitations of the study design were that the 20-g amount of whey protein may have been inadequate to maximally stimulate rates of MPS in all participants and there was no protein-free placebo condition for comparison, making it difficult to discern whether the increased protein synthesis rates were due to the protein ingestion or a result of the contractile-induced stress.

To help answer this question, a subsequent study interrogated the acute effects of protein ingestion on rates of myofibrillar and mitochondrial protein synthesis following a bout of consecutive resistance exercise and cycling in younger (~ 20 years), recreationally trained males [66]. Also utilizing

Table 1 Studies investigating the effects of protein supplementation on rates of muscle protein synthesis or makers of exercise-induced muscle damage and inflammation following concurrent resistance and endurance-based exercise

Study	Participant cohort	Intervention	Protein availability	Main outcomes
Camera et al. [64]	Young, recreationally trained males Age ~ 23 years; $n = 16$ (8/group)	END: 10-min bouts of one-legged cycling @ 75% VO_{2peak} (2 min recovery) until fatigue Power output ↓ by 10 W (same protocol as above until fatigue) 10-min recovery 90 s one-legged max sprints (60 s recovery) until fatigue 30 min arm cranking Overnight recovery REX: unilateral leg press 8 × 5 reps @ 80% 1RM for each leg	2 groups (parallel groups) (1)PRO + CHO: 20 g whey + 40 g maltodextrin consumed immediately post and 2 h post exercise (2)PLA: water + artificial sweetener consumed immediately post and 2 h post exercise	↑ post-exercise myofibrillar FSR PRO + CHO vs PLA ↔ post-exercise myofibrillar FSR between 'high' and 'low' glycogen legs between conditions
Donges et al. [65]	Middle-aged, sedentary males Age ~ 53 years ($n = 8$)	3 conditions (randomized counterbalanced) 1.REX: 8 × 8 reps leg extension @ 70% 1RM 2.END: 40 min cycling @ 55% PPO 3.Concurrent: 4 × 8 reps leg extension @ 70% 1RM; no recovery; 20 min cycling @ 55% PPO	20 g WPI immediately post exercise	↑ post-exercise myofibrillar FSR ($P < 0.05$) with REX (1.8-fold) and concurrent (2.2-fold) Greater ↑ post-exercise myofibrillar FSR with both REX and concurrent compared to END ($P < 0.05$) ↑ post-exercise mitochondrial FSR ($P < 0.05$) with REX (1.3-fold), concurrent (1.5-fold) and END (1.4-fold)
Pasiakos et al. [46]	Physically active (military personnel) women and men Age ~ 21 years ($n = 39$: 7 ♀, 32 ♂)	Concurrent training program designed to elicit 10% energy deficit END: treadmill and cycle ergometry @ 40–60% VO_{2peak} (daily 21 days) REX: 3 × 15 reps one single-joint movement per major muscle group @ pre-determined intensity (3 days/week for 21 days)	3 conditions (randomized cross-over; 30% energy deficiency in all groups for 21 days) 1.0.8 g kg/day RDA 2.1.6 g kg/day 3.2.4 g kg/day	↑ FFM loss post intervention with 0.8 g kg/day RDA compared to 1.6 g kg/day and 2.4 g kg/day groups ↔ FFM loss post intervention between 1.6 g kg/day and 2.4 g kg/day groups ↑ postprandial MPS ($P < 0.05$) in 2.4 g kg/day group only during ED
Camera et al. [66]	Young, recreationally trained males Age ~ 19 years ($n = 8$)	REX: 8 × 5 reps leg extension @ 80% 1RM 15 min recovery END: 30 min cycling @ 70% VO_{2peak}	2 conditions (randomized cross-over) 1.PRO: 25 g whey protein 2.PLA: water + artificial sweetener	↑ post-exercise myofibrillar FSR ($P < 0.05$) with PRO (~ 145%) and PLA (~ 75%) ↑ PRO higher than PLA (~ 67%, $P < 0.05$) ↔ mitochondrial FSR post exercise or between groups

Table 1 (continued)

Study	Participant cohort	Intervention	Protein availability	Main outcomes
Eddens et al. [128]	Young male, well-trained endurance cyclists Age ~ 27 years ($n = 24$; 8/group) VO_{2peak} ~ 61 mL/kg/min	HIIT: 66 sprints of 5, 10 or 15 s duration (work-to-recovery ratio of 1:6, 1:3 or 1:1, respectively) 2 × 2 min and 1 × 5 min sustained repetitions 15 min passive recovery REX: 100 drop-jumps from a height of 0.63 m	3 conditions (parallel groups) 1.PRO: 20 g whey protein hydrolysate (1.85 g CHO; 90 kcal) 2.CHO: 0 g protein, 20.5 g CHO, 90 kcal 3.PLA: 0 g protein, 0 g CHO, 5 kcal All groups consumed supplement twice daily: 3 h preceding, directly following or 3 h following exercise Non-supplemental dietary intake standardized to 6 g/kg/day CHO, 1.2 g/kg/day protein	↓ Maximal voluntary contraction and CMJ post exercise for all groups ($P < 0.05$); no differences between groups ↔ TT time or average power (W/kg) sustained during TT ↑ muscle soreness and plasma CK and CRP post exercise for all groups ($P < 0.05$) No differences in muscle soreness and plasma CK or CRP between groups
Knuiman et al. [105]	Young, recreationally active males Age ~ 21 years ($n = 13$)	END: 90 min cycling @ 70% VO_{2max} Recovery 4 h REX: 5 × 8 reps bilateral leg press + leg extension @ 80% 1RM	2 conditions/meals (randomized crossover) consumed 2 h post END 1.Fat meal: 52 g protein, 51 g fat, 20 g CHO, 3207 kJ 2.CHO meal: 52 g protein, 9 g fat, 110 g CHO, 3124 kJ Both conditions additionally received 10 g casein ~ 10 min after baseline biopsy, 15 g casein immediately post END and 25 g whey protein post REX	↑ plasma IL-6 post exercise with both meals compared to baseline ($P < 0.05$) ↑ IL-6 post REX with fat meal compared to CHO meal ($P < 0.05$) ↑ plasma IL-8 and IL-10 post exercise with both meals compared to baseline ($P < 0.05$) No differences in IL-8 and IL-10 post exercise between meals ↔ plasma TNFα post exercise compared to baseline or between meals
Churchward-Venne et al. [69]	Young, recreationally trained males Age ~ 23 years ($n = 36$; 12/group)	4 × 8 reps leg press @ 80% 1RM + 4 × 8 reps leg extension @ 80% 1RM No recovery 30 min cycling @ ~ 60% W_{Max}	3 conditions (parallel groups) 1.Whey: 45 g CHO + 20 g whey protein (2.58 g Leu) 2.Soy: 45 g CHO + 20 g soy protein (1.44 g Leu) 3.Soy + Leu: 45 g CHO + 20 g soy protein + 2.58 g Leu	↑ post-exercise myofibrillar FSR ($P < 0.05$; main effect of time) with early (0–120 min) vs late (120–360 min) ↔ aggregate (0–360 min) myofibrillar FSR between treatments ↑ post-exercise mitochondrial FSR ($P < 0.05$) with soy + Leu compared to whey and soy only during 0–120 min ↔ aggregate (0–360 min) mitochondrial FSR between treatments

↑ increase/improved performance, ↓ decrease/reduced performance, ↔ no change, ♀ female, ♂ male, *CHO* carbohydrate, *CK* creatine kinase, *CMJ* countermovement jump, *CRP* C reactive protein, *ED* energy deficit, *END* endurance exercise, *FFM* fat free mass, *FSR* fractional synthetic rate, *HIIT* high-intensity interval training, *IL* interleukin, *Leu* leucine, *MPS* muscle protein synthesis, *PLA* placebo, *PPO* peak power output, *REX* resistance exercise, *RM* repetition maximum, *s* second, *TNF* tumor necrosis factor, *TT* time trial, VO_{2peak} peak oxygen uptake, W_{Max} maximal workload, *WPI* whey protein isolate

a cross-over study design, participants performed a resistance exercise session comprising eight sets of five repetitions of leg extension at ~ 80% 1 repetition maximum (1RM) (Table 1). Participants then rested for 15 min before performing 30 min of continuous cycling at a power output that elicited ~ 70% of individual VO_{2peak}. Immediately following the cessation of exercise, participants ingested 500 mL of either an artificial sweetener placebo or 25 g of whey protein. There were selective increases in myofibrillar, but not mitochondrial, protein synthesis rates post exercise in both placebo and protein conditions; however, the increase in the protein condition was significantly higher compared to placebo. Perhaps more importantly, the increase in myofibrillar protein synthesis observed in the protein condition was similar to previous maximal levels observed when protein is ingested following resistance exercise alone [67] despite the performance of endurance exercise in the concurrent exercise bout. This finding provides acute support for post-exercise protein ingestion to stimulate the early muscle protein accretion processes that form the basis for gains in muscle mass over time. Nonetheless, without a resistance exercise-only group, it cannot be discerned from the findings of this study whether the myofibrillar protein synthetic response observed is exclusively the result of the resistance exercise or some interaction with the endurance exercise [68].

Churchward-Venne and co-workers compared the effects of different protein sources, namely whey, soy and soy protein enriched with free leucine (all co-ingested with

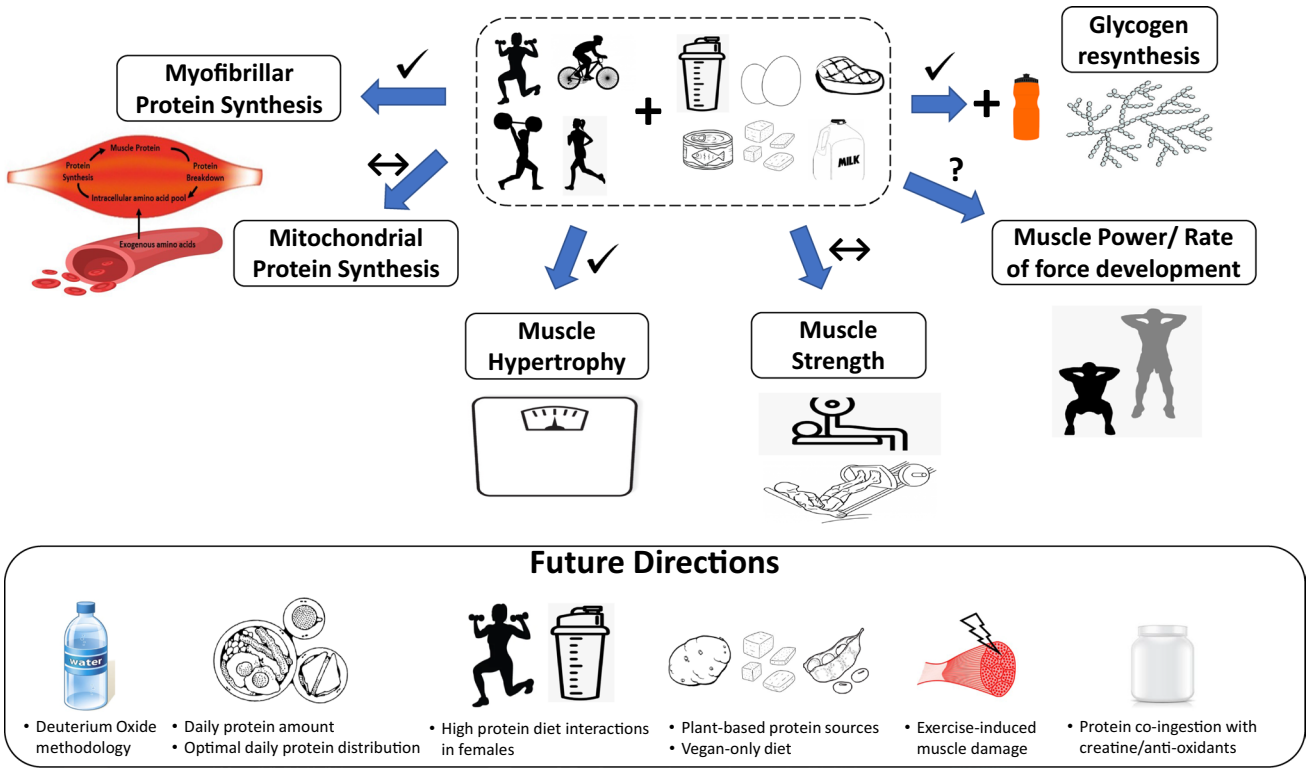

Fig. 1 Protein ingestion following concurrent resistance and aerobic-based exercise stimulates rates of myofibrillar protein synthesis and increased glycogen resynthesis compared to a placebo in the acute recovery (1–5 h) period. In contrast, changes in acute rates of mitochondrial protein synthesis are equivocal. Significant increases in lean body mass have also been reported following longer duration concurrent training programs. Available literature indicates marginal/negligible improvements with increased protein intake on muscle strength and power/rate of force adaptations with longer duration concurrent training programs. Several future directions centered on increased protein intake interactions with concurrent training include the use of deuterium oxide ('heavy water') as a tracer to measure integrated/cumulative rates of MPS for consecutive days and weeks that incorporates the effects of habitual diet, sleeping habits and increased training volume of concurrent training individuals. No study has investigated how to optimize daily protein distribution factoring the likelihood of multiple exercise sessions in a single day; similarly, whether the inherently high training volumes typical of concurrent training programs warrant higher daily protein intakes compared to single-mode training remains an intriguing area of future investigation. In light of some evidence presented for reduced sensitivity of increased protein intake to facilitate anabolic adaptations in females, increased attention needs to be directed at high protein intakes (> 2 g/kg/day) and concurrent training in females. Considering the increasing popularity of plant-based protein diets for ethical and sustainable reasons, how such protein sources modulate adaptive responses with concurrent training is currently unknown. Finally, the magnitude to which exercise-induced muscle damage can alter muscle power-related adaptations with concurrent training, and how creatine co-ingestion with protein may positively regulate these responses in light of its purported increases in power output with exercise, is heavily understudied. *MPS* muscle protein synthesis

carbohydrate) on postprandial rates of myofibrillar and mitochondrial protein synthesis following a single bout of concurrent resistance- and endurance-type exercise in young, healthy, recreationally active men [69] (Table 1). This question stems from differences in the anabolic properties of these protein sources and is of relevance to team-sport environments where some individuals may be (for example) vegan, vegetarian or lactose intolerant and cannot consume animal-based (i.e., whey) protein. Despite greater postprandial plasma leucine concentrations after whey and combined soy and leucine conditions when compared with soy only, similar increases in rates of both myofibrillar and mitochondrial protein synthesis were observed across all protein conditions during the 6 h recovery period [69]. While this study lacked a non-protein condition for comparison, this finding provides initial evidence for individuals participating in concurrent training that post-exercise increases in MPS are comparable between diverse protein sources.

Collectively, findings from these studies and others [46] investigating the effects of protein ingestion with concurrent resistance and aerobic-based exercise provide evidence for the acute stimulation of myofibrillar protein synthesis with protein compared to placebo ingestion. While such increases in MPS may provide meaningful changes in lean body mass with repeated, chronic training programs (Fig. 1), current evidence indicates changes in MPS following a single bout of concurrent exercise with protein intake may not be directly implicated with interference in muscle hypertrophy adaptations with longer duration concurrent training interventions. Nonetheless, the magnitude of MPS response with increased protein intake following concurrent exercise has not been directly compared to resistance exercise performed separately. Furthermore, acute exercise-induced increases in MPS must be extrapolated with caution given previous evidence demonstrating no correlation in such measures with muscle hypertrophy responses following chronic exercise training [70], particularly in studies using traditional stable-isotope tracer methodologies to measure muscle protein fractional synthetic rates. In this regard, no study has incorporated deuterium oxide ('heavy water') as a tracer to measure integrated/cumulative rates of MPS for consecutive days and weeks following a period of concurrent exercise [71]. Such information would provide more 'real-world' measures (i.e., external to the laboratory setting) incorporating cumulative effects of habitual dietary and sleeping habits and, perhaps more importantly, the effects of residual fatigue due to increased exercise training volume that traditional stable-isotope tracers are unable to incorporate. Notably, changes in integrated daily rates of MPS following 3-weeks of resistance training (i.e., coinciding with progressive attenuation of muscle damage) were shown to correlate with muscle hypertrophy responses observed after 10 weeks of training using deuterium oxide tracer methodology [72]. Thus, future studies incorporating this technique may reveal new mechanistic insight into MPS responses with diverse concurrent training programs and protein ingestion strategies that may potentially implicate attenuated MPS responses with previously established long-term interference in muscle strength and power adaptations with concurrent training [35]. It should also be remembered that any acute (minutes to hours) 'interference' at the cellular/molecular level does not necessarily translate into habitual (weeks to months) training responses, particularly in professional sporting settings where multiple matches are played with a short turn-around time, which may compromise nutrition practices [73].

3 Effects of Protein Intake on Lean Body Mass and Muscle Strength with Longer Duration Concurrent Training Interventions

Early work by Antonio and co-workers [74] determining the effects of longer periods of protein ingestion on lean body mass and strength with concurrent training centered on supplementation in the form of the branched-chain amino acids (BCAAs) valine, leucine and isoleucine (Table 2). During this 6-weeks training program in young, untrained females involving whole-body, split-routine resistance training and a combination of cycling, stair stepper and treadmill running, participants received either ~ 18 g of essential amino acids daily or a placebo. No changes in lean body mass or total weight lifted (weight multiplied by repetitions) were reported between or even within groups [74]. The authors postulated this lack of change was related to the short duration of the training program not being able to induce meaningful changes, which is in agreement with a recent systematic review [75] and meta-analysis [58] investigating the effects of protein supplementation on changes in lean mass with exercise. Compounding the short-term training protocol design was the moderately low daily protein intake (~ 1.2 g/kg/day including the 18 g of essential amino acids) of the group consuming the essential amino acids supplement. While this amount is above the current recommended daily allowance for protein, this intake may have been inadequate to promote lean mass accrual with the high training volume performed by participants [76]. Indeed, a very high habitual protein diet (~ 3 g/kg/day) following 6 weeks of concurrent whole-body resistance and rowing training in recreationally active females and males was shown to significantly increase lean body mass along with bench press and leg press maximum strength [77] (Table 2). Additionally, a study by Longland and colleagues in young and overweight (body mass index ~ 29 kg/m^2) males, albeit with a 4-week training duration and ~ 40% caloric deficit (compared to recommended dietary requirements), reported significantly

Table 2 Studies investigating effects of increased protein availability, either through supplementation or a high-protein diet, following short-term (4–12 weeks) and chronic (6 months) concurrent resistance and endurance-based training

Study	Participant Cohort	Intervention	Protein Availability	Main Outcomes
Antonio et al. [74]	Young, untrained females Age ~ 27 years ($n = 19$)	6-week program (3 times/week) REX: split-routine, progressive program with free weights and machines END: 20 min stationary bike, stair stepper or treadmill running @ 70% estimated heart rate max	2 conditions (parallel groups) (1)EAA: 18.3 g EAA (including 3.59 g Leu) ingested daily (split within 20 min pre and post exercise + non-training days) (2)PLA: cellulose (no protein)	↔ body weight or total bone-free lean BM with either group ↔ total weight lifted (weight 3 number of repetitions) with either group ↑ treadmill time to exhaustion only in EAA group ($P < 0.05$)
Ballard et al. [148]	Young, untrained females and males Age ~ 23 years ($n = 51$: 23 ♀, 28 ♂)	6-month program (5 times/week: 3 REX + 2 END odd weeks, 2 REX + 3 END even weeks) REX: whole body program 2 × 10 rep + 1 × AMRAP @ 70% 1RM END: running @ 70% HR_{Max} for 45 min	2 supplement conditions (1)Protein: 42 g protein, 21 g CHO, 1.5 g fat; 280 kcal (2.2 g/kg/day protein throughout intervention) (2)CHO: 70 g CHO, 280 kcal (1.1 g/kg/day protein throughout intervention) Both supplements consumed twice daily: immediately post exercise and in afternoon	↑ FFM with protein compared to CHO at 3 months ($P < 0.05$) but not 6 months ↑ 1RM bench press at 6 months with protein compared to CHO ($P < 0.05$)
Arciero et al. [82]	Young/middle-aged, sedentary females and males Age 26–60 years ($n = 63$: 34 ♀, 29 ♂)	3 exercise groups (parallel groups; 12-week intervention) (1)Concurrent (6 day/week) REX: high-intensity whole-body program including free weights and machines + HIIT: cycling @ ~ 90% RPE for 2–3 mining) (2)Endurance (4–6 day/week): cycling @ 50–75% HR_{Max} for 30–60 min (3)Control: inactive/no exercise	Diet dependent on exercise group allocated to: Concurrent: protein 40% (2.1 g/kg/day), CHO 40%, fat 20% Endurance: protein 10–15% (1.0 g/kg/day), CHO 55–60%, fat < 30%) Control: normal dietary consumption patterns (1.1 g/kg/day protein)	↔ FFM post intervention within or between groups ↑ 1RM post-intervention bench and leg press with concurrent ($P < 0.05$) and compared to endurance and control groups ($P < 0.05$)
Walker et al. [78]	Young, male United States Air Force personnel Age ~ 27 years ($n = 30$)	8-week self-reported END: running 3 days/week REX: push-ups + crunches 0–2 h/week whole-body program	2 conditions (parallel groups) (1)Protein ($n = 18$): 19.7 g of whey protein + 6.2 g Leu (112 kcal) (2)PLA ($n = 12$): 112 kcal CHO (0 g protein) Both conditions consumed respective beverage within ~ 1 h pre and ~ post exercise + non-training days)	↑ bench press 1RM with protein (3.5 kg; 3.9%; $P < 0.05$) and PLA (1.3 kg; 1.4%; $P > 0.05$) ↑ chin-ups ($P < 0.05$) with protein (0.6; 10.1%) vs PLA (0.2; 1.7%) ↔ 3-mile run: 1% improvement with protein vs 3% with PLA ($P > 0.05$) ↔ 40-yard sprint: 4.3% improvement with protein vs 4.8% with PLA ↑ lean body with protein from weeks 1–8 (0.7 ± 1.3 kg) vs ↔ with PLA (0.0 ± 0.9 kg)

Table 2 (continued)

Study	Participant Cohort	Intervention	Protein Availability	Main Outcomes
Longland et al. [54]	Young, recreationally active, overweight males Age: ~ 23 years ($n = 40$; n/20/group) BMI ~ 29 kg/m^2	4-week program (6 days/week) REX: full-body (2 ×/week) circuits composed of 10 reps/set × 3 sets @ 80% 1RM (last set of each exercise to failure) + plyometric body weight circuit (1 ×/week) SIT (1 ×/week): 4–8 30 s Wingate sprints HIIT (1 ×/week): 10 × all-out sprints for 1 min @ 90% PPO + weekly 250-kJ cycle TT time (1 ×/week)	2 conditions (parallel groups) (1)Protein: 2.4 g·kg^{-1}d^{-1} protein, 3.1 CHO g·kg^{-1}d^{-1}, 0.4 g·kg^{-1}d^{-1} fat + weekly dairy WPI supplementation (~ 49 g) (2)Control:1.2 g·kg^{-1}d^{-1} protein, 3.0 CHO g·kg^{-1}d^{-1}, 0.9 g·kg^{-1}d^{-1} fat + weekly dairy WPI supplementation (~ 15 g) Both groups on hypoenergetic diets: ~ 40%reduction compared with dietary requirements	↑ LBM post intervention with protein condition ($P < 0.05$); LBM greater post intervention with protein compared to control ($P < 0.05$) ↓ FM post intervention in both conditions ($P < 0.05$); greater decrease with protein compared to control ($P < 0.05$) ↑ 1RM bench press and leg press 1 post intervention ($P < 0.05$); no differences b/w conditions ↑ VO_{2max} and TT performance post intervention ($P < 0.05$); no differences b/w conditions
Shamim et al. [81]	Recreationally active males Age ~ 27 years ($n = 32$)	3 groups (parallel groups; 12-week intervention) (1)REX: ~ 6–20 repetitions @ 50–90% 1RM whole body program including leg press, knee extension and bench press (3 days/week) (2)END: cycling combination of hill simulation (25–110% of MAP), moderate-intensity continuous cycling @ 50% MAP, moderate-intensity interval training @ 70% MAP and HIIT @ 100% MAP (3 days/week) (3)Concurrent REX and END: same programs as mentioned above performed on alternate days (6 days total/week)	Free-living 2 g/kg/day protein diet 34 g whey protein provided post every exercise session, distribution of protein across 4–6 meals and ~ 20–30 g of protein prior to sleep	↑ total lean mass and vastus lateralis muscle volume post intervention in all groups ($P < 0.05$); no differences between groups ↑ 1RM leg press, leg extension and bench press in concurrent and REX groups post intervention and above END ($P < 0.05$); no differences between REX and concurrent ↑ absolute VO_{2peak} in END and concurrent post intervention ($P < 0.05$); no difference between groups ↑ Wingate peak power relative post intervention with REX greater than concurrent and END ($P = 0.053$)
Ormsbee et al. [93]	Young, recreationally active females and males Age ~ 21 years ($n = 51$: 25 ♀, 26 ♂)	6-month concurrent training program (5 days/week: 3 REX + 2 END odd weeks, 2 REX + 3 END, even weeks) REX: 2 × 10 reps + 1 × AMRAP @ 70% 1RM whole-body exercises END: running (outside or indoor track) 25–45 min @ 70% HR_{Max}	2 conditions (parallel groups) (1)Protein: 42 g protein, 21 g CHO, 1.5 g fat, 280 kcal (~2.2 g/kg/day protein) (2)Control: 0 g protein, 70 g sucrose and fructose; 280 kcal (~1.0 g/kg/day protein) Both supplements consumed 2 times/day: immediately post exercise and 8–12 h post exercise + on non-training days	↑ VO_{2peak} post intervention in both protein and control groups ($P < 0.05$); no differences between groups ↑ 1RM bench press with protein compared to PLA group ($P < 0.05$); no differences between groups in hip sled 1RM ↑ 1RM bench press but not hip sled in women in protein compared to PLA group ($P = 0.1$); ↑ 1RM bench press ($P = 0.03$) and hip sled ($P = 0.07$) in men in protein compared to PLA group ↑ LM post intervention in both protein and control groups ($P < 0.05$); ↑ LM in protein compared to PLA after 3 months ($P < 0.05$)

Table 2 (continued)

Study	Participant Cohort	Intervention	Protein Availability	Main Outcomes
Dicks et al. [79]	Active adults: $n=49$, age ~ 23 years, $n=25$; age ~ 58 years; $n=24$ Sedentary adults ($n=49$: age ~ 26 years, $n=25$; age ~ 57 years; $n=24$)	Self-reported habitual concurrent training activity Sedentary: not participating in REX or END > 1/week Active: participating in > 3/week REX and END @ moderate-to-vigorous intensity	Self-reported (3-day food diary) Protein intake higher ($P<0.05$) in active (~ 1.63 g/kg/day) vs sedentary (~ 1.19 g/kg/day)	Gender, age, concurrent exercise training status + protein intake predicted ($P<0.001$) 71% of variance in muscle functional cross-sectional area
Forbes et al. [77]	Young, recreationally trained, female and males Age ~ 27 years ($n=31$: 15 ♀, 16 ♂)	6-week progressive program REX: 2 days/week, periodized with 6 upper body and 4 lower body exercises @ 60–85% 1RM, 4–12 reps, 2–6 sets END: continuous rowing (machine) 3 days/week @ HR eliciting individual VT + one interval session 1 day/week	3 conditions/supplements (parallel groups) (1)PLA ($n=10$): no protein in supplement; habitual protein intake of 1.2–1.4 g/kg/day reported throughout study (2)WPC ($n=10$): whey protein concentrate (< 80% protein in supplement) (3)WPI ($n=11$): WPI (> 90% protein in supplement) Total habitual protein intake of 3.2 ± 0.3 and 3.8 ± 0.4 g/kg/day for women and men, respectively, reported throughout study	↑ BM, peak VO_{2peak}, 2000-m TT, bench press and leg press 1RM post intervention ($P<0.05$); no differences between groups ↔ white blood cell, neutrophil, lymphocyte and monocyte concentrations post intervention within and between groups
Jendricke et al. [88]	Young, recreationally trained females Age ~ 27 years ($n=59$)	12-week program (3 times/week, > 1-day recovery b/w two sessions) REX: 3 × 20–30 reps squats, lunges, one-legged heel rises END: 1 h running session @ 80–90% velocity at individual anaerobic threshold	2 conditions (parallel groups): (1)Treatment: 15 g specific collagen peptides (7.5 g ingested pre and 7.5 g ingested post exercise) (2)Control: 15 g silicea (silicon dioxide) as PLA	↑ running distance with treatment compared to control ($P<0.05$) ↑ velocity at LT in both treatment and control groups ($P<0.05$); no differences b/w groups ↑ FFM and skeletal muscle mass with treatment above control ($P<0.05$) ↑ 1RM back squats in both treatment and control groups ($P<0.05$) no differences b/w groups

Table 2 (continued)

Study	Participant Cohort	Intervention	Protein Availability	Main Outcomes
Lee et al. [110]	Young, moderately active male Age ~ 25 years ($n = 29$)	3 training groups (parallel groups) (1)REX: 3–4 sets × 12–6 reps @ ~ 70–80% 1RM whole-body program including leg press, bench press, seated row, leg extension, leg curl) (2)Concurrent REX (same as above) + 3 h recovery + HIIT: 8–13 intervals @ 85–97% W_{peak}) (3)Concurrent HIIT (same as above) + 3 h recovery + REX: (same as above) All groups trained 3 days/week for 9 weeks	0.25 g·kg WPI provided immediately after each exercise session Habitual protein intake of ~ 2 g·kg·day reported for all groups	↑ CMJ displacement, velocity, absolute force and power with REX ↑ CMJ peak velocity with HIIT + RT ↓ CMJ displacement, absolute force and power with REX + HIIT compared to REX ↑ 1RM leg press (absolute and relative) in all groups; no differences between groups ↑ total LM and upper body with RT + HIIT ↔ LM and upper body with RT or HIIT + RT; no differences between groups ↑ VO_{2peak} and W_{peak} with RT + HIIT and HIIT + RT (no differences between concurrent groups) ↑ VO_{2peak} and W_{peak} with RT + HIIT and HIIT + RT above RT

↑ increase/improved performance, ↓ decrease/reduced performance, ↔ no change, ♀ female, ♂ male, *AMRAP* as many reps as possible, *BM* body mass, *BMI* body mass index, *CHO* carbohydrate, *CMJ* countermovement jump, *EAA* essential amino acid, *END* endurance exercise, *FFM* fat free mass, *HIIT* high-intensity interval training, HR_{Max} heart rate maximum, *Leu* leucine, *LT* lactate threshold, *MAP* maximum aerobic power, *PLA* placebo, *PPO* peak power output, *REX* resistance exercise, *RM* repetition maximum, *TT* time trial, VO_{2max} maximal oxygen uptake, VO_{2peak} peak oxygen uptake, *VT* ventilatory threshold, W_{peak} aerobic power, *WPI* whey protein isolate

greater post-intervention increases in lean body mass when participants consumed 2.4 g/kg protein ('protein group') per day compared to 1.2 g/day/day ('control group') [54]. A key difference in this work to the previous study by Antonio and co-workers [74] was that participants in this study by Longland and colleagues [54] trained 6 days per week in a program combining whole-body machine/dumbbell and plyometric body circuits concomitant with a mix of sprint interval training, HIT and a weekly time-trial road. This higher resistance training volume may partly explain the significant post-intervention increases in 1RM strength as well as push-up and sit-up performance in both the protein and control groups. Despite these reported increases in lean body mass and strength with protein intervention, neither of these studies included a resistance training-only group to compare the magnitude in responses and thus determine whether interference in such adaptations still exists. Regardless, such findings can still carry implications for individuals who train or perform multiple times per week who, due to their high training volume, may be in a caloric deficit. Promotion of lean body mass during weight loss may also be important in maintaining physical performance while also preserving skeletal muscle (i.e., boxers making weight for an upcoming event or in military training where individuals often undertake concurrent training [78]). Finally, these findings can be of interest more generally for middle-aged adults recreationally engaging in combined resistance and aerobic-based training to improve overall health and fitness while concomitantly attenuating age-related losses in muscle mass [79]. It must be noted that lean body mass was unchanged (i.e., did not decrease) in the control group (protein intake of 1.2 g/day/day) of this study by Longland and colleagues [54]. Thus, if the goal throughout a concurrent training cycle or in-season competition period is to merely maintain lean mass, then protein intakes of ~ 1.2 g/kg/day may be suitable for this purpose while in a caloric deficit.

A limitation of the previously discussed studies is their short-term duration (4–6 weeks). When considering that many of the studies without a nutritional intervention that have investigated the effects of concurrent training do not report 'interference' in muscle strength and hypertrophy adaptations until at least weeks 8–9 of a concurrent training program [27, 31, 32, 80], longer duration interventions would appear to be more suitable to appropriately interrogate the effects of protein ingestion within this training paradigm. Shamim and colleagues were the first to specifically compare the effects of a 12-weeks 'high' protein diet (2 g/kg/day) on changes in lean body mass and strength between groups undertaking resistance or aerobic-based exercise in isolation (each performed 3 days per week) and a concurrent training group (combination of both modalities training 6 days per week) in young and recreationally trained adult males [81] (Table 2). Total lean mass and *vastus lateralis* muscle volume increased post intervention in all groups, a result that may reflect the inclusion of cycling exercise only in the concurrent and endurance-only groups, which can promote leg muscle hypertrophy [68]. However, only the concurrent and resistance-only training groups increased leg press and extension, and bench press strength (relative to body weight). Others have also found superior bench and leg press strength adaptations following 12-weeks of concurrent resistance and endurance training compared to single-mode endurance training (Table 2), although protein intake in this study was twice as high in the concurrent compared to endurance training-only group (2.1 g/kg/day vs 1.0 g/kg/day) [82]. While endurance exercise-induced muscle glycogen depletion can impair muscle strength [83], the absence of strength interference with the concurrent group in the work by Shamim and colleagues may have related to the 24-h recovery period between exercise modalities that, in combination with the carefully planned nutritional interaction, negated post-training residual fatigue. Indeed, this supports previous work [84, 85] that inadequate recovery time/proximity between concurrent exercise sessions can significantly dampen muscle activation and force development that, by extension, can reduce contractile intensity and subsequently dynamic strength gains [86]. Moreover, a meta-analysis of 21 studies, all without planned or reported nutritional (protein) interventions, demonstrated a positive association between duration (length of session) and frequency (days per week) of endurance exercise and the degree of interference with strength gains [35]. Thus, the findings of this study indicate that the potential for attenuated strength adaptations with concurrent training can be alleviated with extended recovery lengths (i.e., 24 h) between concurrent exercise sessions independent of protein intake. Nonetheless, increased protein intake may still represent an important consideration for maintaining strength adaptations with concurrent training when considering the capacity for 'acute' residual fatigue from endurance exercise sessions to compromise the muscle's ability to develop tension during subsequent resistance exercise, which can decrease strength development [87].

More recently, Jendricke and colleagues compared the effects of either of a 15-g collagen peptide supplement or 15 g of silicea placebo on several performance indices following a 12-week concurrent training program incorporating resistance training and running in recreational active female runners [88] (Table 2). Similar increases in back squat 1RM were observed between both the collagen peptide supplement and placebo groups [88]. Interestingly, this comparable increase in 1RM between groups occurred despite the collagen peptide supplement group recording significantly higher increases in muscle mass post intervention, although there was no resistance only training group to compare the magnitude of strength increase. This discordance between changes

in lean body mass and muscle strength contrasts previous findings [89, 90]. However, others have reported disparities in the magnitude of changes in muscle mass and strength with concurrent training [91] and support other available literature that the relationship between exercise training-induced hypertrophy and strength is likely confounded by other variables and may not be purely causal [92].

The capacity for protein intake to augment strength adaptations over longer training periods may be more relevant for both athletes and recreational individuals who undertake periodized training in monthly or in-season micro-cycles. To investigate more chronic effects (i.e., greater than 12 weeks) of protein intake on changes in lean body mass and strength with concurrent training, Ormsbee and coworkers investigated strength and body composition changes to a 6-month concurrent training intervention (Table 2) in which young recreationally active female and male participants consumed either a daily protein (84 g) or carbohydrate (control) supplement of equal caloric value [93]. While changes in lean mass were similar between nutrition conditions across the entire 6-month intervention in females, greater increases in lean mass were reported in males in the protein compared to placebo group after 3 months but not 6 months. Such temporal changes in lean mass with exercise have been reported previously [94]; however, this is the first evidence of divergence between sexes in lean body mass changes with concurrent training. Similar changes were observed for strength adaptation with comparable increases in bench press and hip sled 1RM between nutrition groups in females, while males in the protein condition gained greater strength at both 3- and 6-months time points compared to the control condition [93]. The differences between males and females are difficult to reconcile when considering the protein supplemented group for both sexes consumed twice the amount of protein throughout the intervention compared to the placebo group (~2.2 g/g/day vs ~1.0 g/kg/day). A similar high-protein diet (~2.4/kg/day) was shown to result in significantly greater increases in fat free mass compared to a lower intake (0.9 g/kg/day) in female physique athletes following an 8-week resistance training program [95]. Nonetheless, increases in squat and deadlift 1RM in that study were comparable between these 'high' and 'low' protein groups. This is consistent with other work in untrained females following 12 weeks of whole-body resistance training and milk/dairy supplementation [96], and further highlights both a disparity between changes in lean mass and strength adaptations as well as a negligible effect for protein to augment strength adaptation in females.

Collectively, these findings suggest the capacity for high protein intake to augment increases in lean mass compared to a placebo/low-protein condition with chronic (i.e., 6 months) concurrent training. These increases do not appear to translate into improvements in strength adaptations (Fig. 1), which is congruent with some evidence in resistance training-only studies incorporating high-protein diets [50]. However, it must be remembered that there is a smaller magnitude of interference in muscle hypertrophy compared to strength adaptations with concurrent training [35] that may subsequently increase the likelihood of rescuing interference in hypertrophy responses with increased protein intake. Moreover, the limited number of studies that have directly compared muscle strength and hypertrophy adaptations between concurrent training and resistance training-only groups makes it difficult to ascertain the capacity, if any, for increased protein intake to recover any interference in these adaptation responses. Ensuring appropriate recovery time between adjoining sessions may be more pertinent to maximizing maximal strength responses in a concurrent training setting if permissible [84, 97]. Nonetheless, the potential for increased protein intake to enhance changes in lean mass with concurrent training may be of greater relevance to individuals participating in contact/collision-based sports such as rugby, American Football and Australian Rules Football, where increased muscle mass may assist with holding ground position or tackling/counteracting opposition forward momentum. Moreover, increases in muscle mass are also a priority during preseason training periods for many team sports incorporating concurrent training regimes. More interventional studies combining high-protein diet plans with concurrent training in females are required to better understand the role of protein in anabolic adaptations.

4 Effects of Protein Ingestion on Exercise-Induced Muscle Damage and Associated Muscle Force and Power Output with Concurrent Training

Muscle remodeling processes, as manifested by altered net muscle protein balance, can be elevated above rest for up to 48 h post resistance exercise [98]. Compared to single mode training, individuals participating in high-volume concurrent training incorporating both eccentric muscle actions through resistance exercise and prolonged endurance activity including high-intensity efforts with only minutes or hours between bouts are likely to be susceptible to the potentially unfavorable effects of exercise-induced muscle damage (EIMD) [27]. Repetitive sprint actions with rapid decelerations, characteristic of many performance activities that are incorporated into concurrent training programs, result in eccentric muscle contractions which also contribute to muscle damage [99]. Exercise-induced leakage of intramuscular proteins into the bloodstream, excitation–contraction coupling dysfunction and disruptions to sarcoplasmic cellular calcium homeostasis are characteristic of EIMD [100–102]. The skeletal muscle extracellular matrix is also exposed to significant

degradation and remodeling following muscle-damaging exercise [103]. EIMD presents potential implications for individuals undertaking concurrent training as it has been shown to induce, among a number of responses, increased muscle swelling and soreness, and temporarily attenuate maximal force production [104, 105] and endurance performance [106, 107]. In this regard, muscle power may be the most susceptible adaptation to attenuated improvements with concurrent training [35, 86]. Previous work has demonstrated impaired countermovement jump performance and jump height (measures of maximal muscle force and power output) with concurrent training [33, 35, 80, 108–111], indicating velocity of muscle contraction (i.e., speed at which a muscle changes length) during maximal efforts to be of high importance. In these circumstances when the concurrent training stimulus appears to impair muscle functions for an extended period, there is a justified need to intervene to reduce such negative effects. Approaches including pharmacological (e.g., anti-inflammatory agents), stretching, rehabilitation and physical therapy methods, and nutrition/supplements have been used to alleviate EIMD [112, 113]. When considering that intramuscular remodeling following eccentric-based muscle damage is characterized by mechanical disruption of the sarcolemma [114, 115], addition of new sarcomeres [116, 117] and repair of damaged proteins [117, 118], high protein intake with concurrent training programs may be a strategy to augment such processes and thus accelerate recovery and augment muscle force/power output adaptations.

Previous work investigating the effects of protein intake on EIMD following either a bout of resistance exercise or eccentric contraction-based downhill running has shown protein ingestion can both attenuate decreases in isokinetic muscle performance as well as increase rate of force and power restoration in the 48 h post exercise [119–121]. A recent meta-analysis of eight studies also demonstrated BCAA supplementation can reduce delayed onset muscle soreness following exercise training [122]. Such positive effects of protein/BCAA intake may relate to preserved muscle function with increased protein intake by reducing the magnitude of exercise-induced myofibrillar breakdown [123, 124]. Interestingly, it was recently shown that improved recovery with leucine-enriched amino acids following unaccustomed and damaging resistance exercise did not correlate with changes in MPS [125]. However, such beneficial effects of protein on EIMD-related outcomes are equivocal, potentially due to large variations in study design and biomarkers of muscle damage analyses [55, 126, 127].

Surprisingly, very little is known about the capacity for protein to alter EIMD responses following concurrent training. Such knowledge may assist coaches and sport scientists in designing concurrent training programs that mitigate, where possible, the potential negative effects of EIMD to facilitate recovery. Using a parallel groups design in which participants were provided either 20 g whey protein hydrolysate, carbohydrate or a placebo (both latter conditions containing 0 g protein), Eddens and co-workers investigated blood markers of EIMD and muscle power adaptations following a single bout of concurrent exercise in well-trained endurance cyclists [128] (Table 1). As expected, the bout of concurrent exercise induced substantial EIMD, with reductions in maximal voluntary contraction (MVC) and countermovement jump performance and increases in perceived muscle soreness during the 72-h recovery period. Compared to the carbohydrate and placebo groups, protein supplementation did not attenuate any of the indirect indices of EIMD measured. However, it is difficult to project whether these decrements in MVC and countermovement jump observed with concurrent training would have been of a similar or greater magnitude, without a resistance training-only group for comparison. Indeed, no study has currently shown EIMD responses and their effects on muscle power adaptation responses to be higher with concurrent compared to single-mode resistance training. Regarding the nutritional intervention incorporated throughout the 3-days recovery period of this study, participants consumed a habitual protein intake of 1.2 g/kg/day throughout the recovery period in line with several nutritional positional stands [129]. An important caveat to these findings is that the effects of EIMD were only measured following a single bout of concurrent exercise. Considering skeletal muscle's protective effects against subsequent damaging exercise sessions (i.e., 'repeated bout effect) [130], it is plausible that the decreases in MVC and countermovement jump observed may not be apparent with subsiding EIMD responses with subsequent concurrent training bouts independent of protein intake. It could be postulated that a higher protein intake may be required to either partially or completely offset the acute effects of the EIMD on muscle power/force measurements. Attenuated countermovement displacement, absolute force and muscle power have also been reported following 9 weeks of concurrent resistance and HIIT compared to resistance-only training in recreationally trained males [110] (Table 2). In that study, participants were provided 0.25 g/kg whey protein isolate immediately after each exercise session and had self-reported a habitual protein intake of ~ 2 g/kg/day.

While this attenuation in various indices of muscle force and power with concurrent training supports other previous studies with no nutritional intervention [33, 109, 111], Shamim and colleagues reported no attenuation in squat and countermovement jumps between concurrent and resistance-only groups following 12 weeks of a 2 g/kg/day protein diet [81]. However, this same study did observe reduced improvements in relative Wingate peak power output between concurrent and resistance-only training. No direct measures of EIMD were collected in this study, thereby

making it difficult to associate any potential impairment in Wingate peak power output with elevated intramuscular damage because of the 6-day-a-week concurrent training program. Nonetheless, it is possible that acute reductions in maximal force production previously associated with EIMD [104, 105] may be implicated in the observed interference in Wingate peak power output in the concurrent training group. It is also feasible other mechanisms such as impaired rate of force development [33, 80, 111, 131, 132], changes to fiber type distribution [39] and muscle shortening velocity [133] may play a role in reduced muscle power outputs with long-term concurrent training. The capacity for increased protein intake to augment any of these processes remains a future area of investigation within the concurrent training research field.

When looking at more performance/sporting-based activities, work in canoeists demonstrated increased peak power during arm-cranking as well as rowing time to exhaustion following 6 weeks of daily leucine supplementation (45 mg/kg/day) compared to a placebo, which the authors concluded was due to enhanced recovery and repair of muscle damage from exercise with leucine supplementation [134]. Improvements in high-intensity running and Wingate performance have also been observed with increased protein intake compared to placebo during training programs of soccer players and judoists, respectively [113, 135]. Collectively, these results indicate the potential for increased protein intake to improve muscle power adaptations in select sporting activities requiring concurrent strength, endurance and power for performance. Reduced attenuations during acute recovery in countermovement jump height, knee extensor and flexor peak torque and sprint performance have also been reported with protein supplementation following Australian Rules Football [136], soccer [113] and basketball [137], respectively. The mechanistic underpinning linking increased protein intake with possible improvements in neuromuscular performance has received little scientific scrutiny to date and is an important area for further interrogation.

A notable omission from all longer duration studies that have investigated the effects of protein intake on muscle power and strength outputs with concurrent training is the incorporation of running-based exercise as the aerobic training modality. Running is a core element of many/most individual and team-based activities. As such, even though previous recommendations to avoid a potential interference in adaptation responses with concurrent training stipulate performing cycling-based aerobic exercise instead of running [97], this may not be feasible from a specificity of training point of view if running is a requirement of a given activity. Previous work has demonstrated concurrent running and resistance training can increase running economy and velocity as well as improve performance in a 3-km track running test in male runners [91]. Resistance/strength exercise also can improve neuromuscular characteristics that increase muscle work efficiency and thus allow running-based exercise at a lower oxygen consumption [138–141]. However, compared to cycling, running induces a higher rate of fat oxidation (at the same relative intensity), which is associated with higher total energy expenditure [142]. This in turn can potentiate states of negative energy balance when running sessions are performed repeatedly (i.e.: on the same day or multiple sessions within a week) or in combination with resistance exercise. Running is also more biomechanically dissimilar than cycling to most resistance/weight loading exercise [143]. More pressingly, the weight-bearing, eccentric loading and subsequent muscle damage associated with intense running may induce greater physiological stress (at least compared to cycling) as well as acute fatigue that can collectively impede recovery and thus negatively affect muscle power and force-generating adaptations with concurrent training [35]. Future studies incorporating running-based exercise into concurrent training study designs are required to better elucidate whether increased protein intake can improve and/or rescue potential interference in muscle strength and power adaptations in such training programs.

Another potential avenue to alleviating potential decrements in muscle force/power output with concurrent training is through the co-ingestion of protein with creatine. Numerous studies have ascertained that increased skeletal muscle creatine content via creatine supplementation can improve resistance to fatigue, enhance power output during short maximal sprints, particularly with repeated exercise bouts, and may reduce muscle damage by increasing cell membrane stabilization following intense exercise [144, 145]. Acute creatine supplementation (7 days) was previously shown to prevent impairments in maximal leg and bench press strength following concurrent exercise involving resistance exercise with either running or cycling; however, muscle power and force-generation adaptations were not measured [146]. As phosphocreatine levels decrease with high-intensity exercise [147], long-term creatine supplementation concomitant with a high-protein diet may positively alter previously identified interference in muscle power and strength with concurrent training, although the mechanistic basis for this and whether such a response is synergistic in nature remains an area of future investigation.

In summary, the available evidence indicates that protein intake up to levels twice the recommended intake and beyond does not rescue decrements in selective aspects of muscle force and power production when resistance training is performed concurrently with aerobic-based exercise as opposed to in isolation (Fig. 1). This lends support to findings from a recent meta-analysis concluding that protein supplementation imparts only moderate effects on attenuating muscle peak force and torque losses in the 1–4 days following resistance exercise [148]. In line with the concept

of movement specificity, concurrent training programs that prioritize the consistent performance of more plyometric/power-based exercises that closely model movements to be performed in competition may be required to alleviate some of the previously observed reductions in muscle force and power [139]. Such training strategies would be pertinent to individuals training for activities in which muscle power is central to performance. High protein intake combined with creatine may work to reduce the effects of EIMD on muscle and force-generating adaptations with concurrent training, although this has not yet been investigated. Finally, the roles of central and/or peripheral neuromuscular fatigue that may result in reduced neural activation of skeletal muscles and voluntary force production [149] cannot be discounted as other possible causes of diminished muscle power adaptations with concurrent training compared to resistance exercise performed in isolation.

5 Conclusion

Concurrent training is an effective strategy to simultaneously improve skeletal muscle size, strength, power and oxidative capacity. However, the majority of studies investigating both molecular and physiological adaptation responses to concurrent training have not included nutritional/protein-based interventions which are known to enhance exercise-mediated muscle mass and strength adaptations. Considering the dual exercise modalities with concurrent training and potential for inherently greater training volumes than single-mode exercise training, requirements for greater protein intake in this training exist beyond just stimulating MPS. Specifically, protein is imperative to further promoting the capacity for individuals undertaking concurrent training to recover and train consistently while healthy for optimal performance and continued adaptation. Every hour between concurrent training sessions and competition periods represents a crucial time of ongoing recovery in which nutrition should be a key focus, particularly with increased risks of overtraining due to high training duration and frequency. Future recommendations/guidelines for optimal protein ingestion strategies within a concurrent training program require relevant evidence-based scientific studies factoring the practicalities and constraints present within such a training setting. In this regard, such recommendations should be dependent on the type of exercise components constituting the training program, as well as the duration and intensity of these exercise sessions. Further investigation of protein interactions with concurrent training programs possessing diverse training variables (e.g., exercise mode, frequency, intensity, volume, between-session recovery duration and exercise order) can provide evidence-based information for sports coaches, scientists and practitioners for prescribing concurrent training strategies capable of maximizing muscle strength, power, hypertrophy and aerobic adaptation responses.

Declarations

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of Interest Donny Camera declares that he has no conflicts of interest relevant to the content of this review.

Authorship Contributions DMC wrote and approved the final manuscript.

References

1. Hawley JA, Hargreaves M, Joyner MJ, Zierath JR. Integrative biology of exercise. Cell. 2014;159(4):738–49.
2. Coffey VG, Hawley JA. The molecular bases of training adaptation. Sports Med (Auckland, NZ). 2007;37(9):737–63.
3. Camera DM, Smiles WJ, Hawley JA. Exercise-induced skeletal muscle signaling pathways and human athletic performance. Free Radic Biol Med. 2016;98:131–43.
4. Tanaka H, Swensen T. Impact of resistance training on endurance performance. A new form of cross-training? Sports Med (Auckland, NZ). 1998;25(3):191–200.
5. Wilkinson SB, Phillips SM, Atherton PJ, Patel R, Yarasheski KE, Tarnopolsky MA, et al. Differential effects of resistance and endurance exercise in the fed state on signalling molecule phosphorylation and protein synthesis in human muscle. J Physiol. 2008;586(15):3701–17.
6. Holloszy JO. Biochemical adaptations in muscle. Effects of exercise on mitochondrial oxygen uptake and respiratory enzyme activity in skeletal muscle. J Biol Chem. 1967;242(9):2278–82.
7. Henriksson J, Reitman JS. Time course of changes in human skeletal muscle succinate dehydrogenase and cytochrome oxidase activities and maximal oxygen uptake with physical activity and inactivity. Acta Physiol Scand. 1977;99(1):91–7.
8. Hawley JA. Adaptations of skeletal muscle to prolonged, intense endurance training. Clin Exp Pharmacol Physiol. 2002;29(3):218–22.
9. D'Antona G, Lanfranconi F, Pellegrino MA, Brocca L, Adami R, Rossi R, et al. Skeletal muscle hypertrophy and structure and function of skeletal muscle fibres in male body builders. J Physiol. 2006;570(Pt 3):611–27.
10. Phillips SM, Tipton KD, Ferrando AA, Wolfe RR. Resistance training reduces the acute exercise-induced increase in muscle protein turnover. Am J Physiol. 1999;276(1 Pt 1):E118–24.
11. Zierath JR, Hawley JA. Skeletal muscle fiber type: influence on contractile and metabolic properties. PLoS Biol. 2004;2(10):e348.
12. Harber MP, Konopka AR, Douglass MD, Minchev K, Kaminsky LA, Trappe TA, et al. Aerobic exercise training improves whole muscle and single myofiber size and function in older women. Am J Physiol Regul Integr Comp Physiol. 2009;297(5):R1452–9.
13. Fry CS, Noehren B, Mula J, Ubele MF, Westgate PM, Kern PA, et al. Fibre type-specific satellite cell response to aerobic training in sedentary adults. J Physiol. 2014;592(12):2625–35.
14. Pesta D, Hoppel F, Macek C, Messner H, Faulhaber M, Kobel C, et al. Similar qualitative and quantitative changes of mitochondrial respiration following strength and endurance training in

normoxia and hypoxia in sedentary humans. Am J Physiol Regul Integr Comp Physiol. 2011;301(4):R1078–87.
15. Balakrishnan VS, Rao M, Menon V, Gordon PL, Pilichowska M, Castaneda F, et al. Resistance training increases muscle mitochondrial biogenesis in patients with chronic kidney disease. Clin J Am Soc Nephrol. 2010;5(6):996–1002.
16. Gibala M. Molecular responses to high-intensity interval exercise. Appl Physiol Nutr Metab. 2009;34(3):428–32.
17. Gibala MJ, Little JP, Macdonald MJ, Hawley JA. Physiological adaptations to low-volume, high-intensity interval training in health and disease. J Physiol. 2012;590(5):1077–84.
18. McMillan K, Helgerud J, Macdonald R, Hoff J. Physiological adaptations to soccer specific endurance training in professional youth soccer players. Br J Sports Med. 2005;39(5):273–7.
19. Wong PL, Chaouachi A, Chamari K, Dellal A, Wisloff U. Effect of preseason concurrent muscular strength and high-intensity interval training in professional soccer players. J Strength Cond Res. 2010;24(3):653–60.
20. Bell KE, Seguin C, Parise G, Baker SK, Phillips SM. Day-to-Day changes in muscle protein synthesis in recovery from resistance, aerobic, and high-intensity interval exercise in older men. J Gerontol A Biol Sci Med Sci. 2015;70(8):1024–9.
21. Callahan MJ, Parr EB, Hawley JA, Camera DM. Can high-intensity interval training promote skeletal muscle anabolism? Sports Med. 2021;51(3):405–21.
22. World Health Organization. Global recommendations on physical activity for health. 2010.
23. Balabinis CP, Psarakis CH, Moukas M, Vassiliou MP, Behrakis PK. Early phase changes by concurrent endurance and strength training. J Strength Cond Res. 2003;17(2):393–401.
24. Schumann M, Yli-Peltola K, Abbiss CR, Häkkinen K. Cardiorespiratory adaptations during concurrent aerobic and strength training in men and women. PLoS ONE. 2015;10(9): e0139279.
25. Pedersen BK, Saltin B. Exercise as medicine—evidence for prescribing exercise as therapy in 26 different chronic diseases. Scand J Med Sci Sports. 2015;25(Suppl 3):1–72.
26. Coffey VG, Hawley JA. Concurrent exercise training: Do opposites distract? J Physiol. 2017;595(9):2883–96.
27. Fyfe JJ, Bishop DJ, Stepto NK. Interference between concurrent resistance and endurance exercise: molecular bases and the role of individual training variables. Sports Med (Auckland, NZ). 2014;44(6):743–62.
28. Mohr M, Krustrup P, Bangsbo J. Match performance of high-standard soccer players with special reference to development of fatigue. J Sports Sci. 2003;21(7):519–28.
29. Wisløff U, Castagna C, Helgerud J, Jones R, Hoff J. Strong correlation of maximal squat strength with sprint performance and vertical jump height in elite soccer players. Br J Sports Med. 2004;38(3):285–8.
30. Jenner SL, Buckley GL, Belski R, Devlin BL, Forsyth AK. Dietary Intakes of professional and semi-professional team sport athletes do not meet sport nutrition recommendations—a systematic literature review. Nutrients. 2019;11(5):1160.
31. Bell GJ, Syrotuik D, Martin TP, Burnham R, Quinney HA. Effect of concurrent strength and endurance training on skeletal muscle properties and hormone concentrations in humans. Eur J Appl Physiol. 2000;81(5):418–27.
32 Dolezal BA, Potteiger JA. Concurrent resistance and endurance training influence basal metabolic rate in nondieting individuals. J Appl Physiol (Bethesda, Md: 1985). 1998;85(2):695–700.
33. Fyfe JJ, Bartlett JD, Hanson ED, Stepto NK, Bishop DJ. Endurance training intensity does not mediate interference to maximal lower-body strength gain during short-term concurrent training. Front Physiol. 2016;7:487.
34. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. Eur J Appl Physiol. 1980;45(2–3):255–63.
35. Wilson JM, Marin PJ, Rhea MR, Wilson SM, Loenneke JP, Anderson JC. Concurrent training: a meta-analysis examining interference of aerobic and resistance exercises. J Strength Cond Res. 2012;26(8):2293–307.
36. Ahtiainen JP, Hulmi JJ, Kraemer WJ, Lehti M, Pakarinen A, Mero AA, et al. Strength, [corrected] endurance or combined training elicit diverse skeletal muscle myosin heavy chain isoform proportion but unaltered androgen receptor concentration in older men. Int J Sports Med. 2009;30(12):879–87.
37. McCarthy JP, Pozniak MA, Agre JC. Neuromuscular adaptations to concurrent strength and endurance training. Med Sci Sports Exerc. 2002;34(3):511–9.
38 Methenitis S. A brief review on concurrent training: from laboratory to the field. Sports (Basel, Switzerland). 2018;6(4):127.
39. Kazior Z, Willis SJ, Moberg M, Apró W, Calbet JA, Holmberg HC, et al. Endurance exercise enhances the effect of strength training on muscle fiber size and protein expression of Akt and mTOR. PLoS ONE. 2016;11(2): e0149082.
40 Lundberg TR, Fernandez-Gonzalo R, Tesch PA. Exercise-induced AMPK activation does not interfere with muscle hypertrophy in response to resistance training in men. J Appl Physiol (Bethesda, Md: 1985). 2014;116(6):611–20.
41. Doma K, Deakin GB, Schumann M, Bentley DJ. Training considerations for optimising endurance development: an alternate concurrent training perspective. Sports Med (Auckland, NZ). 2019;49(5):669–82.
42 Hawley JA, Burke LM, Phillips SM, Spriet LL. Nutritional modulation of training-induced skeletal muscle adaptations. J Appl Physiol (Bethesda, Md: 1985). 2011;110(3):834–45.
43. Lopez-Miranda J, Williams C, Lairon D. Dietary, physiological, genetic and pathological influences on postprandial lipid metabolism. Br J Nutr. 2007;98(3):458–73.
44. Burke LM, Kiens B, Ivy JL. Carbohydrates and fat for training and recovery. J Sports Sci. 2004;22(1):15–30.
45. Areta JL, Burke LM, Camera DM, West DW, Crawshay S, Moore DR, et al. Reduced resting skeletal muscle protein synthesis is rescued by resistance exercise and protein ingestion following short-term energy deficit. Am J Physiol Endocrinol Metab. 2014;306(8):E989–97.
46. Pasiakos SM, Cao JJ, Margolis LM, Sauter ER, Whigham LD, McClung JP, et al. Effects of high-protein diets on fat-free mass and muscle protein synthesis following weight loss: a randomized controlled trial. FASEB J. 2013;27(9):3837–47.
47. Burke LM, Hawley JA, Wong SH, Jeukendrup AE. Carbohydrates for training and competition. J Sports Sci. 2011;29(Suppl 1):S17-27.
48. Perez-Schindler J, Hamilton DL, Moore DR, Baar K, Philp A. Nutritional strategies to support concurrent training. Eur J Sport Sci. 2015;15(1):41–52.
49. Cintineo HP, Arent MA, Antonio J, Arent SM. Effects of protein supplementation on performance and recovery in resistance and endurance training. Front Nutr. 2018;5:83.
50. Jäger R, Kerksick CM, Campbell BI, Cribb PJ, Wells SD, Skwiat TM, et al. International Society of Sports Nutrition Position Stand: protein and exercise. J Int Soc Sports Nutr. 2017;14:20.
51. Kato H, Suzuki K, Bannai M, Moore DR. Protein requirements are elevated in endurance athletes after exercise as determined by the indicator amino acid oxidation method. PLoS ONE. 2016;11(6): e0157406.
52 Lemon PW, Tarnopolsky MA, MacDougall JD, Atkinson SA. Protein requirements and muscle mass/strength changes during intensive training in novice bodybuilders. J Appl Physiol (Bethesda, Md: 1985). 1992;73(2):767–75.

53. Phillips SM. Dietary protein requirements and adaptive advantages in athletes. Br J Nutr. 2012;108(Suppl 2):S158–67.
54. Longland TM, Oikawa SY, Mitchell CJ, Devries MC, Phillips SM. Higher compared with lower dietary protein during an energy deficit combined with intense exercise promotes greater lean mass gain and fat mass loss: a randomized trial. Am J Clin Nutr. 2016;103(3):738–46.
55. Pasiakos SM, Lieberman HR, McLellan TM. Effects of protein supplements on muscle damage, soreness and recovery of muscle function and physical performance: a systematic review. Sports Med (Auckland, NZ). 2014;44(5):655–70.
56 Burd NA, Tang JE, Moore DR, Phillips SM. Exercise training and protein metabolism: influences of contraction, protein intake, and sex-based differences. J aAppl Physiol (Bethesda, Md: 1985). 2009;106(5):1692–701.
57. Phillips SM, Hartman JW, Wilkinson SB. Dietary protein to support anabolism with resistance exercise in young men. J Am Coll Nutr. 2005;24(2):134s-s139.
58. Cermak NM, Res PT, de Groot LC, Saris WH, van Loon LJ. Protein supplementation augments the adaptive response of skeletal muscle to resistance-type exercise training: a meta-analysis. Am J Clin Nutr. 2012;96(6):1454–64.
59. Morton RW, Murphy KT, McKellar SR, Schoenfeld BJ, Henselmans M, Helms E, Aragon AA, Devries MC, Banfield L, Krieger JW, Phillips SM. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. Br J Sports Med. 2018;52(6):376–84.
60. Breen L, Philp A, Witard OC, Jackman SR, Selby A, Smith K, et al. The influence of carbohydrate-protein co-ingestion following endurance exercise on myofibrillar and mitochondrial protein synthesis. J Physiol. 2011;589(Pt 16):4011–25.
61. Coffey VG, Moore DR, Burd NA, Rerecich T, Stellingwerff T, Garnham AP, et al. Nutrient provision increases signalling and protein synthesis in human skeletal muscle after repeated sprints. Eur J Appl Physiol. 2011;111(7):1473–83.
62. Harber MP, Konopka AR, Jemiolo B, Trappe SW, Trappe TA, Reidy PT. Muscle protein synthesis and gene expression during recovery from aerobic exercise in the fasted and fed states. Am J Physiol Regul Integr Comp Physiol. 2010;299(5):R1254–62.
63. Lunn WR, Pasiakos SM, Colletto MR, Karfonta KE, Carbone JW, Anderson JM, et al. Chocolate milk and endurance exercise recovery: protein balance, glycogen, and performance. Med Sci Sports Exerc. 2012;44(4):682–91.
64 Camera DM, West DW, Burd NA, Phillips SM, Garnham AP, Hawley JA, et al. Low muscle glycogen concentration does not suppress the anabolic response to resistance exercise. J Appl Physiol (Bethesda, Md: 1985). 2012;113(2):206–14.
65 Donges CE, Burd NA, Duffield R, Smith GC, West DW, Short MJ, et al. Concurrent resistance and aerobic exercise stimulates both myofibrillar and mitochondrial protein synthesis in sedentary middle-aged men. J Appl Physiol (Bethesda, Md: 1985). 2012;112(12):1992–2001.
66. Camera DM, West DW, Phillips SM, Rerecich T, Stellingwerff T, Hawley JA, et al. Protein ingestion increases myofibrillar protein synthesis after concurrent exercise. Med Sci Sports Exerc. 2015;47(1):82–91.
67. Moore DR, Tang JE, Burd NA, Rerecich T, Tarnopolsky MA, Phillips SM. Differential stimulation of myofibrillar and sarcoplasmic protein synthesis with protein ingestion at rest and after resistance exercise. J Physiol. 2009;587(Pt 4):897–904.
68. Konopka AR, Harber MP. Skeletal muscle hypertrophy after aerobic exercise training. Exerc Sport Sci Rev. 2014;42(2):53–61.
69. Churchward-Venne TA, Pinckaers PJM, Smeets JSJ, Peeters WM, Zorenc AH, Schierbeek H, et al. Myofibrillar and mitochondrial protein synthesis rates do not differ in young men following the ingestion of carbohydrate with whey, soy, or leucine-enriched soy protein after concurrent resistance- and endurance-type exercise. J Nutr. 2019;149(2):210–20.
70. Mitchell CJ, Churchward-Venne TA, Parise G, Bellamy L, Baker SK, Smith K, et al. Acute post-exercise myofibrillar protein synthesis is not correlated with resistance training-induced muscle hypertrophy in young men. PLoS ONE. 2014;9(2): e89431.
71. Camera DM, Burniston JG, Pogson MA, Smiles WJ, Hawley JA. Dynamic proteome profiling of individual proteins in human skeletal muscle after a high-fat diet and resistance exercise. FASEB J. 2017;31(12):5478–94.
72. Damas F, Phillips SM, Libardi CA, Vechin FC, Lixandrão ME, Jannig PR, et al. Resistance training-induced changes in integrated myofibrillar protein synthesis are related to hypertrophy only after attenuation of muscle damage. J Physiol. 2016;594(18):5209–22.
73. Anderson L, Orme P, Naughton RJ, Close GL, Milsom J, Rydings D, et al. Energy intake and expenditure of professional soccer players of the english premier league: evidence of carbohydrate periodization. Int J Sport Nutr Exerc Metab. 2017;27(3):228–38.
74 Antonio J, Sanders MS, Ehler LA, Uelmen J, Raether JB, Stout JR. Effects of exercise training and amino-acid supplementation on body composition and physical performance in untrained women. Nutrition (Burbank, Los Angeles County, Calif). 2000;16(11–12):1043–6.
75. Pasiakos SM, McLellan TM, Lieberman HR. The effects of protein supplements on muscle mass, strength, and aerobic and anaerobic power in healthy adults: a systematic review. Sports Med (Auckland, NZ). 2015;45(1):111–31.
76. Lemon PW. Effects of exercise on dietary protein requirements. Int J Sport Nutr. 1998;8(4):426–47.
77. Forbes SC, Bell GJ. Whey protein isolate or concentrate combined with concurrent training does not augment performance, cardiorespiratory fitness, or strength adaptations. J Sports Med Phys Fitness. 2020;60(6):832–40.
78. Walker TB, Smith J, Herrera M, Lebegue B, Pinchak A, Fischer J. The influence of 8 weeks of whey-protein and leucine supplementation on physical and cognitive performance. Int J Sport Nutr Exerc Metab. 2010;20(5):409–17.
79. Dicks ND, Kotarsky CJ, Trautman KA, Barry AM, Keith JF, Mitchell S, et al. Contribution of protein intake and concurrent exercise to skeletal muscle quality with aging. J Frailty Aging. 2020;9(1):51–6.
80. Häkkinen K, Alen M, Kraemer WJ, Gorostiaga E, Izquierdo M, Rusko H, et al. Neuromuscular adaptations during concurrent strength and endurance training versus strength training. Eur J Appl Physiol. 2003;89(1):42–52.
81. Shamim B, Devlin BL, Timmins RG, Tofari P, Lee Dow C, Coffey VG, et al. Adaptations to concurrent training in combination with high protein availability: a comparative trial in healthy, recreationally active men. Sports Med (Auckland, NZ). 2018;48(12):2869–83.
82. Arciero PJ, Gentile CL, Martin-Pressman R, Ormsbee MJ, Everett M, Zwicky L, et al. Increased dietary protein and combined high intensity aerobic and resistance exercise improves body fat distribution and cardiovascular risk factors. Int J Sport Nutr Exerc Metab. 2006;16(4):373–92.
83. Jacobs I, Kaiser P, Tesch P. Muscle strength and fatigue after selective glycogen depletion in human skeletal muscle fibers. Eur J Appl Physiol. 1981;46(1):47–53.
84. Robineau J, Babault N, Piscione J, Lacome M, Bigard AX. Specific training effects of concurrent aerobic and strength exercises depend on recovery duration. J Strength Cond Res. 2016;30(3):672–83.

85. Sporer BC, Wenger HA. Effects of aerobic exercise on strength performance following various periods of recovery. J Strength Cond Res. 2003;17(4):638–44.
86. Eddens L, van Someren K, Howatson G. The role of intra-session exercise sequence in the interference effect: a systematic review with meta-analysis. Sports Med (Auckland, NZ). 2018;48(1):177–88.
87. Craig BW, Lucas J, Pohlman R, Stelling H. The effects of running, weightlifting and a combination of both on growth hormone release. J Strength Condition Res. 1991;5(4):198–203.
88. Jendricke P, Kohl J, Centner C, Gollhofer A, König D. Influence of specific collagen peptides and concurrent training on cardiometabolic parameters and performance indices in women: a randomized controlled trial. Front Nutr. 2020;7: 580918.
89. Maughan RJ, Watson JS, Weir J. Strength and cross-sectional area of human skeletal muscle. J Physiol. 1983;338:37–49.
90. Narici MV, Landoni L, Minetti AE. Assessment of human knee extensor muscles stress from in vivo physiological cross-sectional area and strength measurements. Eur J Appl Physiol. 1992;65(5):438–44.
91. Sedano S, Marín PJ, Cuadrado G, Redondo JC. Concurrent training in elite male runners: the influence of strength versus muscular endurance training on performance outcomes. J Strength Cond Res. 2013;27(9):2433–43.
92. Nuzzo JL, Finn HT, Herbert RD. Causal mediation analysis could resolve whether training-induced increases in muscle strength are mediated by muscle hypertrophy. Sports Med (Auckland, NZ). 2019;49(9):1309–15.
93. Ormsbee MJ, Willingham BD, Marchant T, Binkley TL, Specker BL, Vukovich MD. Protein supplementation during a 6-month concurrent training program: effect on body composition and muscular strength in sedentary individuals. Int J Sport Nutr Exerc Metab. 2018;28(6):619–28.
94. Churchward-Venne TA, Tieland M, Verdijk LB, Leenders M, Dirks ML, de Groot LC, et al. There are no nonresponders to resistance-type exercise training in older men and women. J Am Med Dir Assoc. 2015;16(5):400–11.
95. Campbell BI, Aguilar D, Conlin L, Vargas A, Schoenfeld BJ, Corson A, et al. Effects of high versus low protein intake on body composition and maximal strength in aspiring female physique athletes engaging in an 8-week resistance training program. Int J Sport Nutr Exerc Metab. 2018;28(6):580–5.
96. Josse AR, Tang JE, Tarnopolsky MA, Phillips SM. Body composition and strength changes in women with milk and resistance exercise. Med Sci Sports Exerc. 2010;42(6):1122–30.
97. Murach KA, Bagley JR. Skeletal muscle hypertrophy with concurrent exercise training: contrary evidence for an interference effect. Sports Med (Auckland, NZ). 2016;46(8):1029–39.
98. Phillips SM, Tipton KD, Aarsland A, Wolf SE, Wolfe RR. Mixed muscle protein synthesis and breakdown after resistance exercise in humans. Am J Physiol. 1997;273(1 Pt 1):E99-107.
99. Naughton M, Miller J, Slater GJ. Impact-induced muscle damage: performance implications in response to a novel collision simulator and associated timeline of recovery. J Sports Sci Med. 2018;17(3):417–25.
100. Gissel H, Clausen T. Excitation-induced Ca^{2+} influx and skeletal muscle cell damage. Acta Physiol Scand. 2001;171(3):327–34.
101. Peake JM, Neubauer O, Della Gatta PA, Nosaka K. Muscle damage and inflammation during recovery from exercise. J Appl Physiol (Bethesda, Md: 1985). 2017;122(3):559–70.
102. Takarada Y. Evaluation of muscle damage after a rugby match with special reference to tackle plays. Br J Sports Med. 2003;37(5):416–9.
103. Hyldahl RD, Nelson B, Xin L, Welling T, Groscost L, Hubal MJ, et al. Extracellular matrix remodeling and its contribution to protective adaptation following lengthening contractions in human muscle. FASEB J. 2015;29(7):2894–904.
104. Howatson G, van Someren KA. The prevention and treatment of exercise-induced muscle damage. Sports Med (Auckland, NZ). 2008;38(6):483–503.
105. Knuiman P, Hopman MTE, Hangelbroek R, Mensink M. Plasma cytokine responses to resistance exercise with different nutrient availability on a concurrent exercise day in trained healthy males. Physiol Rep. 2018;6(11): e13708.
106. Braun WA, Dutto DJ. The effects of a single bout of downhill running and ensuing delayed onset of muscle soreness on running economy performed 48 h later. Eur J Appl Physiol. 2003;90(1–2):29–34.
107. Chen TC, Nosaka K, Lin MJ, Chen HL, Wu CJ. Changes in running economy at different intensities following downhill running. J Sports Sci. 2009;27(11):1137–44.
108. Kraemer WJ, Patton JF, Gordon SE, Harman EA, Deschenes MR, Reynolds K, et al. Compatibility of high-intensity strength and endurance training on hormonal and skeletal muscle adaptations. J Appl Physiol (Bethesda, Md: 1985). 1995;78(3):976–89.
109. Chtara M, Chaouachi A, Levin GT, Chaouachi M, Chamari K, Amri M, et al. Effect of concurrent endurance and circuit resistance training sequence on muscular strength and power development. J Strength Cond Res. 2008;22(4):1037–45.
110. Lee MJ, Ballantyne JK, Chagolla J, Hopkins WG, Fyfe JJ, Phillips SM, et al. Order of same-day concurrent training influences some indices of power development, but not strength, lean mass, or aerobic fitness in healthy, moderately-active men after 9 weeks of training. PLoS ONE. 2020;15(5): e0233134.
111. Rønnestad BR, Hansen EA, Raastad T. High volume of endurance training impairs adaptations to 12 weeks of strength training in well-trained endurance athletes. Eur J Appl Physiol. 2012;112(4):1457–66.
112. Minett GM, Costello JT. Specificity and context in post-exercise recovery: it is not a one-size-fits-all approach. Front Physiol. 2015;6:130.
113. Poulios A, Fatouros IG, Mohr M, Draganidis DK, Deli C, Papanikolaou K, et al. Post-game high protein intake may improve recovery of football-specific performance during a congested game fixture: results from the PRO-FOOTBALL study. Nutrients. 2018;10(4):494.
114. Morgan DL. New insights into the behavior of muscle during active lengthening. Biophys J. 1990;57(2):209–21.
115. Proske U, Morgan DL. Muscle damage from eccentric exercise: mechanism, mechanical signs, adaptation and clinical applications. J Physiol. 2001;537(Pt 2):333–45.
116. Bongiovanni T, Genovesi F, Nemmer M, Carling C, Alberti G, Howatson G. Nutritional interventions for reducing the signs and symptoms of exercise-induced muscle damage and accelerate recovery in athletes: current knowledge, practical application and future perspectives. Eur J Appl Physiol. 2020;120(9):1965–96.
117. Yu JG, Carlsson L, Thornell LE. Evidence for myofibril remodeling as opposed to myofibril damage in human muscles with DOMS: an ultrastructural and immunoelectron microscopic study. Histochem Cell Biol. 2004;121(3):219–27.
118. Yu JG, Fürst DO, Thornell LE. The mode of myofibril remodelling in human skeletal muscle affected by DOMS induced by eccentric contractions. Histochem Cell Biol. 2003;119(5):383–93.
119 Cockburn E, Hayes PR, French DN, Stevenson E, St Clair Gibson A. Acute milk-based protein-CHO supplementation attenuates exercise-induced muscle damage. Appl Physiol Nutr Metab. 2008;33(4):775–83.
120. Etheridge T, Philp A, Watt PW. A single protein meal increases recovery of muscle function following an acute eccentric exercise bout. Appl Physiol Nutr Metab. 2008;33(3):483–8.

121. Hoffman JR, Ratamess NA, Tranchina CP, Rashti SL, Kang J, Faigenbaum AD. Effect of a proprietary protein supplement on recovery indices following resistance exercise in strength/power athletes. Amino Acids. 2010;38(3):771–8.
122. Fedewa MV, Spencer SO, Williams TD, Becker ZE, Fuqua CA. Effect of branched-chain amino acid supplementation on muscle soreness following exercise: a meta-analysis. Int J Vitamin Nutr Res. 2019;89(5–6):348–56.
123. Howatson G, Hoad M, Goodall S, Tallent J, Bell PG, French DN. Exercise-induced muscle damage is reduced in resistance-trained males by branched chain amino acids: a randomized, double-blind, placebo controlled study. J Int Soc Sports Nutr. 2012;9:20.
124. Shimomura Y, Yamamoto Y, Bajotto G, Sato J, Murakami T, Shimomura N, et al. Nutraceutical effects of branched-chain amino acids on skeletal muscle. J Nutr. 2006;136(2):529s-s532.
125. Waskiw-Ford M, Hannaian S, Duncan J, Kato H, Abou Sawan S, Locke M, et al. Leucine-enriched essential amino acids improve recovery from post-exercise muscle damage independent of increases in integrated myofibrillar protein synthesis in young men. Nutrients. 2020;12(4):1061.
126. Ferguson-Stegall L, McCleave EL, Ding Z, Doerner PG 3rd, Wang B, Liao YH, et al. Postexercise carbohydrate-protein supplementation improves subsequent exercise performance and intracellular signaling for protein synthesis. J Strength Cond Res. 2011;25(5):1210–24.
127. Poulios A, Georgakouli K, Draganidis D, Deli CK, Tsimeas PD, Chatzinikolaou A, et al. protein-based supplementation to enhance recovery in team sports: What is the evidence? J Sports Sci Med. 2019;18(3):523–36.
128. Eddens L, Browne S, Stevenson EJ, Sanderson B, van Someren K, Howatson G. The efficacy of protein supplementation during recovery from muscle-damaging concurrent exercise. Appl Physiol Nutr Metab. 2017;42(7):716–24.
129. Thomas DT, Erdman KA, Burke LM. Position of the academy of nutrition and dietetics, Dietitians of Canada, and the American college of sports medicine: nutrition and athletic performance. J Acad Nutr Diet. 2016;116(3):501–28.
130. Hyldahl RD, Chen TC, Nosaka K. Mechanisms and mediators of the skeletal muscle repeated bout effect. Exerc Sport Sci Rev. 2017;45(1):24–33.
131 Dudley GA, Djamil R. Incompatibility of endurance- and strength-training modes of exercise. J Appl Physiol (Bethesda, Md: 1985). 1985;59(5):1446–51.
132. Mikkola J, Rusko H, Izquierdo M, Gorostiaga EM, Häkkinen K. Neuromuscular and cardiovascular adaptations during concurrent strength and endurance training in untrained men. Int J Sports Med. 2012;33(9):702–10.
133. Linari M, Bottinelli R, Pellegrino MA, Reconditi M, Reggiani C, Lombardi V. The mechanism of the force response to stretch in human skinned muscle fibres with different myosin isoforms. J Physiol. 2004;554(Pt 2):335–52.
134. Crowe MJ, Weatherson JN, Bowden BF. Effects of dietary leucine supplementation on exercise performance. Eur J Appl Physiol. 2006;97(6):664–72.
135. Laskowski R, Antosiewicz J. Increased adaptability of young judo sportsmen after protein supplementation. J Sports Med Phys Fitness. 2003;43(3):342–6.
136. Lee NA, Fell JW, Pitchford NW, Hall AH, Leveritt MD, Kitic CM. Combined carbohydrate and protein ingestion during Australian rules football matches and training sessions does not reduce fatigue or accelerate recovery throughout a weeklong junior tournament. J Strength Cond Res. 2018;32(2):344–55.
137. Gentle HL, Love TD, Howe AS, Black KE. A randomised trial of pre-exercise meal composition on performance and muscle damage in well-trained basketball players. J Int Soc Sports Nutr. 2014;11:33.
138. Ferrauti A, Bergermann M, Fernandez-Fernandez J. Effects of a concurrent strength and endurance training on running performance and running economy in recreational marathon runners. J Strength Cond Res. 2010;24(10):2770–8.
139. Jung AP. The impact of resistance training on distance running performance. Sports Med (Auckland, NZ). 2003;33(7):539–52.
140. Millet GP, Jaouen B, Borrani F, Candau R. Effects of concurrent endurance and strength training on running economy and VO(2) kinetics. Med Sci Sports Exercise. 2002;34(8):1351–9.
141. Sunde A, Støren O, Bjerkaas M, Larsen MH, Hoff J, Helgerud J. Maximal strength training improves cycling economy in competitive cyclists. J Strength Cond Res. 2010;24(8):2157–65.
142 Achten J, Jeukendrup AE. Optimizing fat oxidation through exercise and diet. Nutrition (Burbank, Los Angeles County, Calif). 2004;20(7–8):716–27.
143. Gergley JC. Comparison of two lower-body modes of endurance training on lower-body strength development while concurrently training. J Strength Cond Res. 2009;23(3):979–87.
144. Gualano B, Roschel H, Lancha AH Jr, Brightbill CE, Rawson ES. In sickness and in health: the widespread application of creatine supplementation. Amino Acids. 2012;43(2):519–29.
145. Kreider RB, Kalman DS, Antonio J, Ziegenfuss TN, Wildman R, Collins R, et al. International Society of Sports Nutrition position stand: safety and efficacy of creatine supplementation in exercise, sport, and medicine. J Int Soc Sports Nutr. 2017;14:18.
146. de Salles PV, Alves VT, Ugrinowitsch C, Benatti FB, Artioli GG, Lancha AH Jr, et al. Creatine supplementation prevents acute strength loss induced by concurrent exercise. Eur J Appl Physiol. 2014;114(8):1749–55.
147. Krustrup P, Mohr M, Steensberg A, Bencke J, Kjaer M, Bangsbo J. Muscle and blood metabolites during a soccer game: implications for sprint performance. Med Sci Sports Exerc. 2006;38(6):1165–74.
148 Davies RW, Carson BP, Jakeman PM. The effect of whey protein supplementation on the temporal recovery of muscle function following resistance training: a systematic review and meta-analysis. Nutrients. 2018;10(2):221.
149 Carroll TJ, Taylor JL, Gandevia SC. Recovery of central and peripheral neuromuscular fatigue after exercise. J Appl Physiol (Bethesda, Md: 1985). 2017;122(5):1068–76.